

Ain Shams University
Faculty of Engineering
I-Credit Hours Engineering Programs

Semester: Spring 2026

Course Code: CSE 481 and CSE 371

Project Description

Project Objectives

By the end of these projects, students should be able to apply the theoretical concepts they learned in the course:

1. Construct a closed loop feedback control system for position control systems.
2. The effect of the PID controller parameters on the system performance.
3. The tuning of the controller parameters.

Project Description

The aim of this project is to design a position control system for a geared DC motor. The system should be able to achieve any angular position set point in the range of 0 to 180. The aim of the project will be used in an experimental setup for applying the theoretical knowledge gained in a step-by-step manner throughout the stages of the project.

Layouts

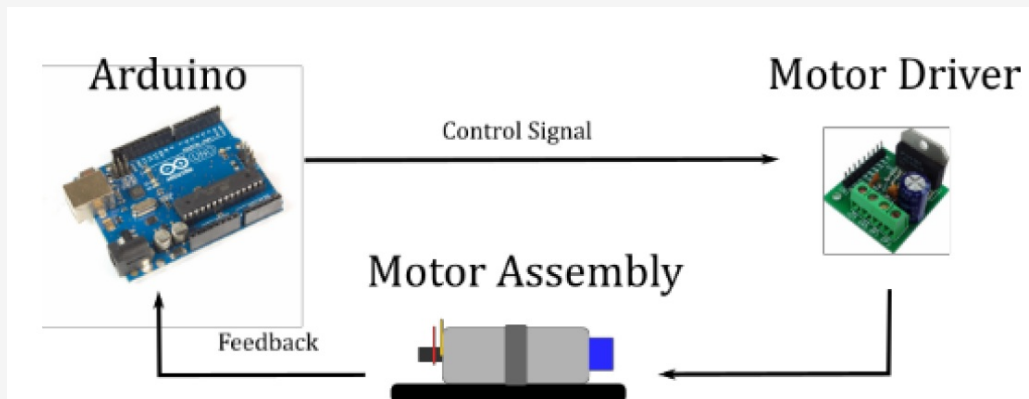


Figure 1: **Position Control project layout**

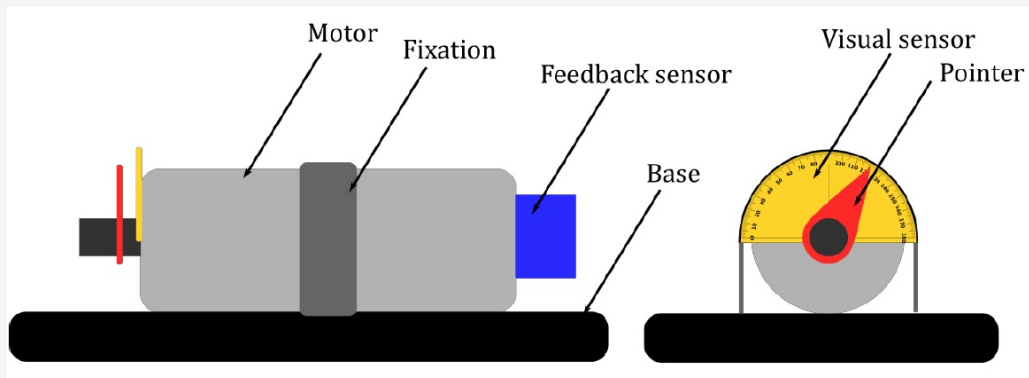


Figure 2: **Motor Assembly**

The project setup consists of a geared DC motor. Attached to the motor, an angular position sensor for feedback. Figure 1: Position Control project layout Figure 1 and Figure 2 show the complete layout of the project. The controller will be implemented using the Arduino microcontroller. The Arduino will be programmed so that it can read the feedback angle from the potentiometer and sends out a PWM (pulse-width modulation) signal, based on the PID control law, which supplies this signal as an output to the motor driver which controls the speed and direction of the motor until it reaches the required position.

Components

- Geared DC Motor.
- Microcontroller (Arduino).
- Motor driver.
- Angular Position sensor (Potentiometer Encoder).
- Implementation of visual sensor (protractor) and pointer.

Requirements

You should construct the hardware system. The motor must be fixed well with a pointer attached to its shaft and moves with it, you should also design and apply a visual sensor to measure the angle of the pointer, such as the (protractor).

You should implement PID Controller on Arduino to control the angle of the pointer and make it move to the desired angle.

You should submit a report contains the following:

1. The objective of the project and how it works.
2. Block diagram of the closed loop control system, showing the components and the flow of the signals between these components.
3. Describe the control action, final control element (actuator), sensor, and controlled variable in your system.
4. A Photo shows your overall real hardware setup (the complete system as shown in the setup as you can).
5. Photo of each component and its name and function.
6. Four graphs from your trials to tune the system, each graph should plot the controlled variable with time as a given set-point. For each graph mentions the PID parameters (K_p , K_i and K_d) used in that trial, and for each graph mention the most suitable PID parameters from your point of view. You also have to mention the PID parameters, steady state error, rise time, overshoot and settle time.
7. A graph represents the final response of the system (the controlled variable with time) after tuning and choosing the most suitable PID parameters from your trials at certain set point. Where your trial should have at most two peaks and not oscillating till settling down (You could choose not only one point, but four or three at once for better picture).
8. The complete code.

You will work in groups of max. 5 students.

Deadline: Week 14

Marking Rubric

Assessment Component	Weight
Presentation	20%
Oral Discussion	20%
Hardware Performance	30%
Simulation using MATLAB	20%
Report	10%

End of Project Description

Good luck with your project!